$$1 + r^1 + r^2 + r^3 + \dots + r^n = \frac{1 - r^{n+1}}{1 - r}$$

Proof By Well Ordering Principle, $P(n) = \frac{1 - r^{n+1}}{1 - r}$

$$C ::= \{ n \in \mathbb{N} \mid P(n) \text{ is false} \}$$

By Well Ordering Principle, there will be smallest element, $*C$.

since $P(0) = 1 = \frac{1 - r}{1 - r} = 1 \Rightarrow$ True

therfore $C > 0$, meaning $(C-1)$ is non-negative integer that still true

$$1 + r^1 + r^2 + r^3 + \dots + r^{(C-1)} = \frac{1 - r^C}{1 - r}$$

add r^C both side

$$1 + r^1 + \dots + r^{(C-1)} + r^C = \frac{1 - r^C}{1 - r} + r^C$$

$$= \frac{1 - r^C + r^C - r^{C+1}}{1 - r}$$

$$= \frac{1 - r^{C+1}}{1 - r}$$

which means that formula does hold for C. This is a contradiction ∎

1.

$$1+r^1+r^2+\cdots+r^n = \frac{1-r^{n+1}}{1-r} \qquad r \neq 1$$

Proof by induction

$$\text{let } P(n) = \frac{1-r^{n+1}}{1-r}$$

Base case

$$P(0) = 1 = \frac{1-r^1}{1-r} \Rightarrow \text{True}$$

Induction step

for $n \geq 0$, show $P(n) \Rightarrow P(n+1)$ is True $n \in N$

Assume $P(n)$ is True, for the purpose of induction

$$1+r^1+r^2+\cdots+r^n+r^{n+1} = \frac{1-r^{n+1}}{1-r} + r^{n+1} \quad (\text{By } P(n))$$

$$= \frac{1-r^{n+1}+r^{n+1}-r^{n+2}}{1-r}$$

$$= \frac{1-r^{n+2}}{1-r}$$

This show $P(n+1)$ is True.

so $P(n) \Rightarrow P(n+1) \quad \forall n \geq 0$

By induction principle, $P(n)$ is true for all $n \geq 1 \quad n \in N$

(2)

purple-eye contestant number : p

red-eye contestant number : r

$p \geq 1 \quad r \geq 0$

$p \geq 1$ ⇒ There is at least on purple-eye .

n = number of day after announcement including announcement day = 1

let's think

if there is only $p = 1$,

he/she would see everyone's head is red meaing his/her is purpule so he/she will leave that day = 1 ($n = 1$)

if $p = 2$, there is jack and Mike who is purple

both Jack and Mike will see each other's purple eye and assume another will leave first day

both neither of them leave at first day.

Jack think "Mike don't leave b.c he saw a another purple eye b.c if he didn't see, $p = 1$ statement will follow but other people expect Mike & Me are red. Mike is purple and he saw my purple and thinking I will leave I am purple I must leave

Vice visa for Mike

they both will leave at $n = 2$.

if $p = 3$; Jack, Mike and Merry.

$n = 1$, no one will leave same as ($p = 2$)

$n = 2$, no one leave thinking other two will leave.

b.c it they were only two ($p = 2$) pattern will follow

$n = 3$, Merry think "if $p = 2$, they both will leave at $n = 2$, second day"

↓

assume merry is red

"but no one leave b.c there are $p \geq 2$, but every one except Jack, Mike is red. so my head is purple"

All of three think the same at $n = 3$ at left that day

we will use this predicate

$P(n) = \begin{cases} \text{if } p > n \Rightarrow \text{no one will leave} \\ \text{if } p = n \Rightarrow \text{all purple eye people leave} \\ \text{if } p < n \Rightarrow \text{only red purple eye poeple left in island} \end{cases}$

Proof by induction

$P(n) = \begin{cases} \text{if } p > n \Rightarrow \text{no one will leave} \\ \text{if } p = n \Rightarrow \text{all purple eye people leave} \\ \text{if } p < n \Rightarrow \text{only red purple eye people is left in island} \end{cases}$

Base case

$P(1) = p > 1 \Rightarrow$ no one leave accord to previous state True

$p = 1 \Rightarrow$ he/she know everyone is red. so he is purple and leave True

$p < 1 \Rightarrow$ since $p \geq 1$, $p < 1$ is false. false $\Rightarrow$ T/F is True

Induction step

for $n \geq 1$. show $P(n) \Rightarrow P(n+1)$ is true

Assume $P(n)$ is true for the purpose of induction

Case A: $p > n+1$

$p > n+1 \Rightarrow p > n$

since people are not leaving n days b.c $p > n$ b.c they see $p > n$ eye

if $p > n+1$, since each purple contestant see at least $n+1$ purple eye and they don't leave if $p > n$, so they would also leave if $p > n+1$

Case B: $p = n+1$

$p = n+1 \Rightarrow p > n$

since p purple people are not leaving at n day. each of them know $p > n$ which can be $p = n+1$ (or) $p > n+1$. If each of them see n purple eye, each know $p > n$. he know he have purple eye and leave at $n+1$.

Case C: $p < n+1$

two possible sub case

1. $p = n$. since purple leaved at day n. there will be no purpule people at island
2. $p < n$: they have already left island so there will be no purple people at island

$P(n) \Rightarrow P(n+1)$ $n \geq 0$

By the principle of induction $P(n)$ is true for all $n \geq 0$.